

Executive Memo: Demand Forecasting & Inventory Planning

To: Director of Retail Operations

From: Lavanya Kaushik, Data Analyst

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Subject: Better forecasts can free €4–6M of inventory and sharpen promo spend

Bottom line. A machine-learning forecast (LightGBM) cuts weekly store-level forecast error by 44% compared with the current seasonal-naïve method. Used for replenishment, it lets our 1,115 stores hold 37–54% less safety stock at the same 95% service target. The promotion analysis also points to a shorter promo that is worth testing.

What we found

1. Forecast accuracy. Across a four-window rolling backtest (2014–15), forecast error (WAPE) fell from 11.2% to 6.3%. LightGBM also beat Excel's FORECAST.ETS on sample stores (4.9% vs 6.5%). The biggest gains come when the promo calendar shifts, which the seasonal method cannot follow. The 8 summer/tourist stores remain the hardest to forecast (8.9% error).

2. Inventory. At a 95% service target, safety stock falls from €19.7M to €9.0M in sales value. Tested on data the policy had not seen, it delivered 97.8% actual service, and at the same achieved service level the reduction is 37%. That frees **€4.4–6.4M of inventory at cost**, or about €0.9–1.3M a year in holding cost.

3. Promotions. Promo days lift sales by 37% (95% CI 36–38%), from both more shoppers (+19%) and bigger baskets (+15%). The lift is front-loaded: +65% on Monday but only +18% on Friday. A Monday–Wednesday promo would keep about 75% of the promo sales gain with 60% of the promo days. The 25 summer/tourist and Sunday-trading stores get about half the average lift, and stores 789 and 794 show none.

Recommendations

- 1 Pilot forecast-based replenishment** at a 95% service target in one region, tracking stockouts and inventory weekly against a comparison region. Chain-wide potential: €4.4–6.4M of inventory freed.
- 2 Run the Monday–Wednesday promo test** that has already been designed: 2 promo weeks, 225 test stores vs 890 control stores, matched by store cluster and size. It reliably detects a 2% change in sales. Adopt the shorter promo only if the sales lost are smaller than the promo cost saved.
- 3 Review promo participation** for the 25 low-response stores and for stores 789 and 794.

Caveats and next steps

The data (Rossmann, 2013–15) records sales in euros, not units, so inventory is measured in sales value. Cost figures assume a 60% cost of goods and a 20% annual holding cost, lead time is assumed to be one week, and promo costs are not in the data. Next steps: separate holiday and regular-season safety stock, and better forecasts for summer/tourist stores.

Explore the numbers in the interactive [Tableau dashboard](#) and [what-if app](#).